
Algorithm 6: Label-setting algorithm with time-budgeted UB pruning, scoring, and early stopping

In this initial version, we use the following scoring formula:

$$\text{Score}(L) = |C| + (1 - \tau)UB; \text{ where } \tau = \frac{t - \text{start_time}}{1440}$$

Input:

- V . Set of nodes
- E_s . Set of scheduled edges (v, u, dep, arr)
- E_t . Set of transfer edges (v, w, Δ)
- $\text{country}(v)$. Country associated with node v
- $R(v, b)$. Precomputed reachable country sets
- $MTT(c)$. Precomputed minimum transit times for each country
- $\text{bucket}(t)$. Time bucket mapping function
- record . Current record to beat (e.g. 13)

Output:

- P . Sequence of labels representing a route within 24 hours

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1: Initialize Labels( $v$ )  $\leftarrow \emptyset$  for all  $v \in V$ 
2: Initialize priority queue  $Q \leftarrow \emptyset$  // ordered by decreasing score
3: Initialize best_label  $\leftarrow$  null
4: // Initialize labels at departures
5: for all  $v \in V$  do
6:   for all scheduled edges  $(v, u, dep, arr) \in E_s$  do
7:      $L \leftarrow (u, arr, \{\text{country}(u)\}, \text{pred} = \text{null}, \text{start\_time} = dep)$ 
8:     Labels( $u$ )  $\leftarrow$  Labels( $u$ )  $\cup \{L\}$ 
9:     compute score( $L$ )
10:     $Q \leftarrow Q \cup \{L\}$ 
11: // Main loop
12: while  $Q \neq \emptyset$  do
13:   extract  $L = (v, t, C, \text{pred}, \text{start\_time})$  with maximum score from  $Q$ 
14:   // Enforce 24-hour journey limit (1440 absolute minutes)
15:   if  $t - \text{start\_time} > 1440$  then
16:     continue
17:   // Upper bound computation (Time-Budgeted)
18:    $b \leftarrow \text{bucket}(t)$ 
19:    $U \leftarrow R(v, b) \setminus C$ 
20:   Sort  $U$  as a sequence  $(c_1, c_2, \dots, c_k)$  such that  $MTT(c_1) \leq MTT(c_2) \leq \dots \leq MTT(c_k)$ 
21:   time_spent  $\leftarrow 0$ 
22:   UB  $\leftarrow 0$ 
23:   remaining_time  $\leftarrow 1440 - (t - \text{start\_time})$ 
24:   for  $i = 1$  to  $|U|$  do
25:     time_spent  $\leftarrow$  time_spent +  $MTT(c_i)$ 
26:     if time_spent  $\leq$  remaining_time then
27:       UB  $\leftarrow$  UB + 1
28:     else
29:       break
30:   // Record-based pruning
31:   if  $|C| + \text{UB} \leq \text{record}$  then
32:     continue
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33: // Update best label
34: if best_label = null  $\vee$   $|C| > |\text{best\_label}.C|$  then
35:   best_label  $\leftarrow L$ 
36:   if  $|C| > \text{record}$  then
37:     break // early stopping: record beaten
38: // Extend via scheduled edges
39: for all  $(v, u, \text{dep}, \text{arr}) \in E_s$  do
40:   if  $\text{dep} \geq t \wedge \text{dep} - \text{start\_time} \leq 1440$  then
41:      $t_{\text{new}} \leftarrow \text{arr}$ 
42:     if  $t_{\text{new}} - \text{start\_time} > 1440$  then
43:       continue
44:      $C' \leftarrow C \cup \{\text{country}(u)\}$ 
45:      $L' \leftarrow (u, t_{\text{new}}, C', \text{pred} = L, \text{start\_time} = \text{start\_time})$ 
46:     if  $\nexists L'' \in \text{Labels}(u)$  such that
47:        $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C' \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C'))$  then
48:       remove all  $L'' \in \text{Labels}(u)$  such that
49:          $(t_{\text{new}} \leq L''.t \wedge C' \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C' \neq L''.C))$ 
50:        $\text{Labels}(u) \leftarrow \text{Labels}(u) \cup \{L'\}$ 
51:       compute score( $L'$ )
52:        $Q \leftarrow Q \cup \{L'\}$ 
53: // Extend via transfers
54: for all  $(v, w, \Delta) \in E_t$  do
55:    $t_{\text{new}} \leftarrow t + \Delta$ 
56:   if  $t_{\text{new}} - \text{start\_time} > 1440$  then
57:     continue
58:    $L' \leftarrow (w, t_{\text{new}}, C, \text{pred} = L, \text{start\_time} = \text{start\_time})$ 
59:   if  $\nexists L'' \in \text{Labels}(w)$  such that
60:      $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C))$  then
61:     remove all  $L'' \in \text{Labels}(w)$  such that
62:        $(t_{\text{new}} \leq L''.t \wedge C \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C \neq L''.C))$ 
63:      $\text{Labels}(w) \leftarrow \text{Labels}(w) \cup \{L'\}$ 
64:     compute score( $L'$ )
65:      $Q \leftarrow Q \cup \{L'\}$ 
66: // Reconstruct path
67:  $P \leftarrow$  empty sequence
68:  $L \leftarrow \text{best\_label}$ 
69: while  $L \neq \text{null}$  do
70:   prepend  $L$  to  $P$ 
71:    $L \leftarrow L.\text{pred}$ 
72: return  $P$ 

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In Algorithm 6 we incorporate the MTT precomputation (see Algorithm 5) in our label-setting algorithm to try and accelerate convergence and pruning of unpromising solutions.